

# Monument Scavenger Hunt Game

Team: Stress, Math and AI

SMAI Assignment 3

## Problem Statement

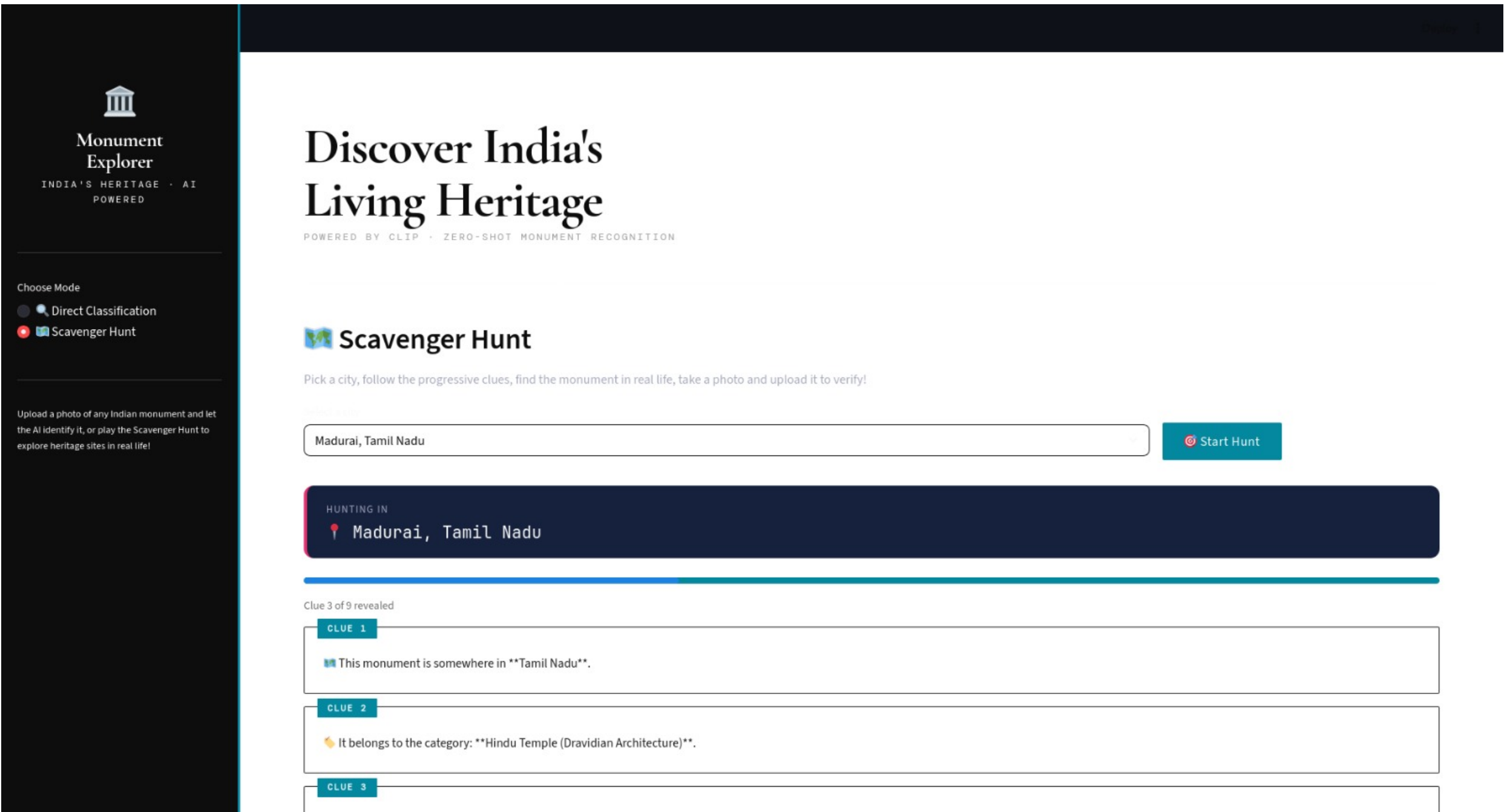
Exploring historical monuments often requires tourists to switch between disconnected search engines, maps, and static information pages, interrupting immersion and reducing engagement during on-site exploration.

Existing monument-recognition systems primarily focus on passive image identification and fail to provide interactive or context-aware exploration experiences. They typically lack:

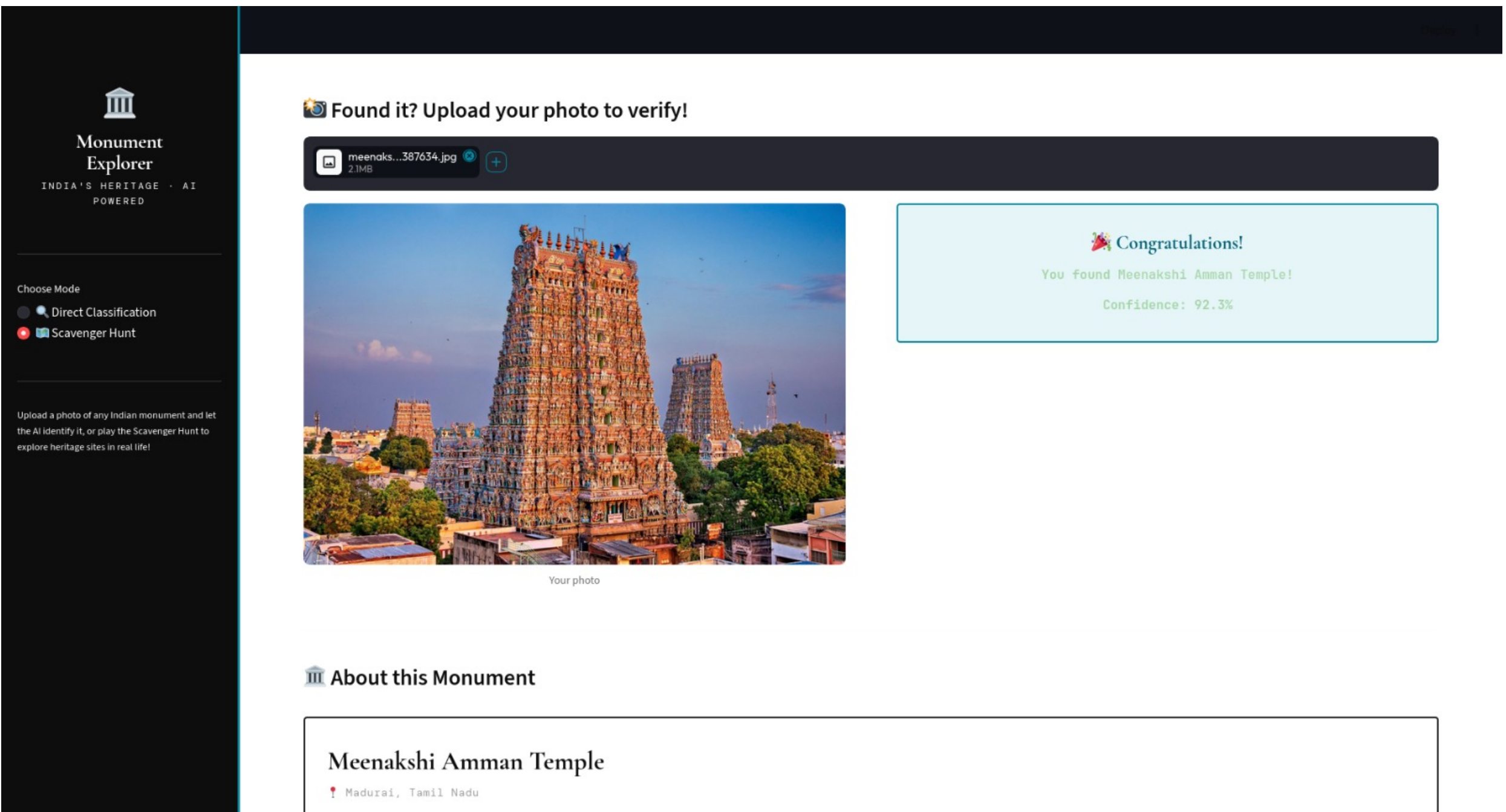
- Gamified and immersive monument discovery
- Real-time monument verification from user photographs
- Contextual historical and travel information
- Interactive clue-based exploration workflows

This motivates the need for an AI-assisted heritage exploration system that combines monument verification, gamified scavenger hunts, and contextual information retrieval within a unified in-teractive experience.

## Interactive Exploration Workflow



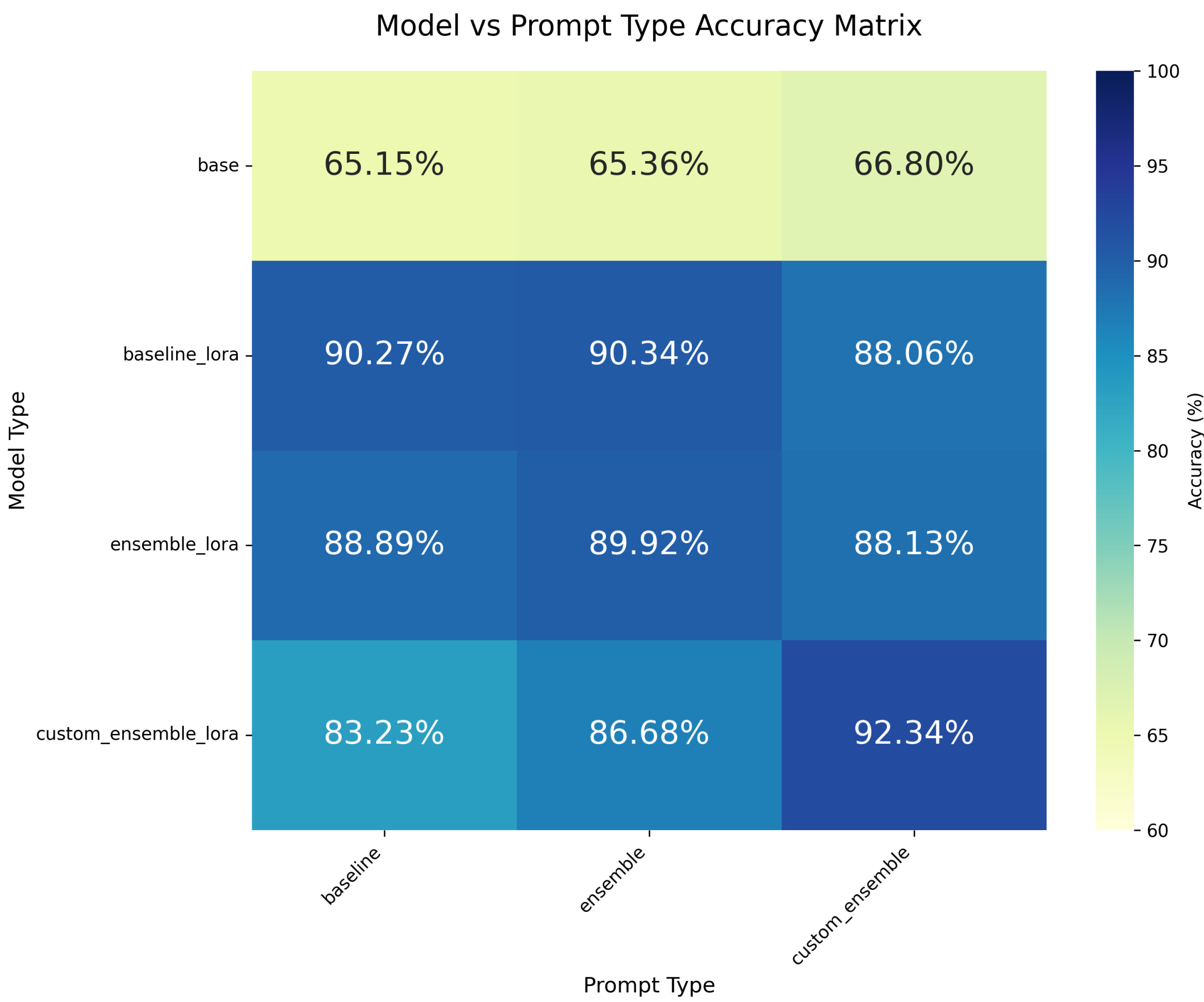
Scavenger Hunt Interface



AI-Based Monument Verification

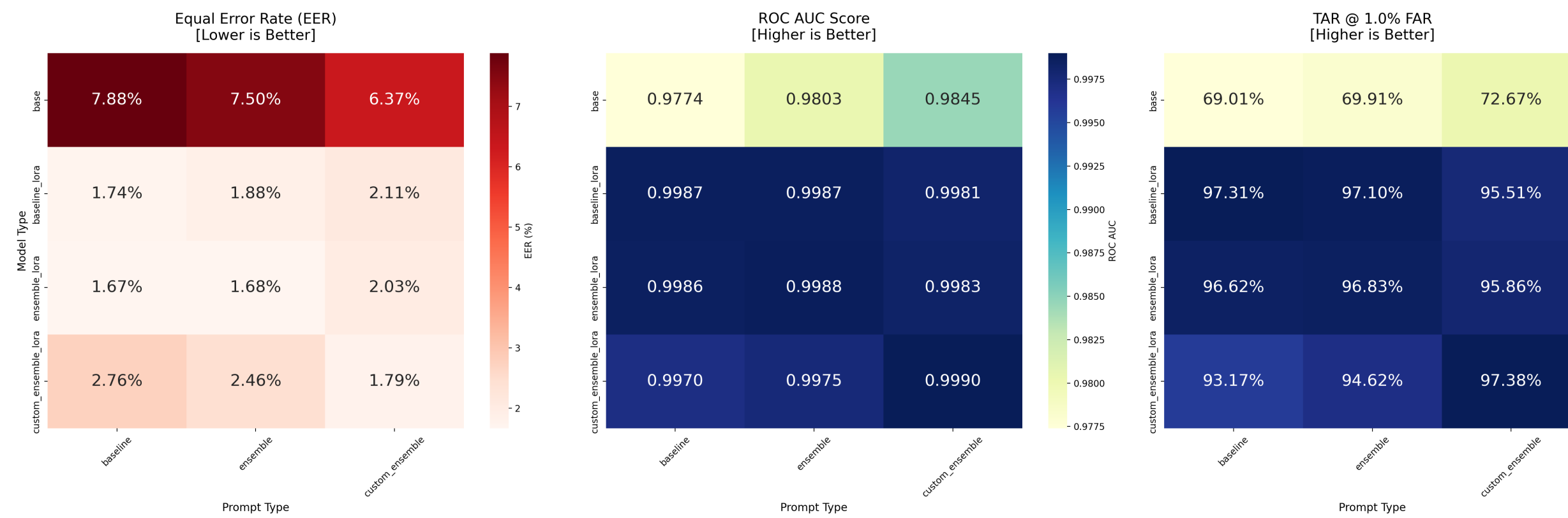
The application introduces a gamified monument exploration workflow where users progressively solve cryptic clues to identify real-world heritage sites. Users upload monument photographs through the Streamlit interface, after which CLIP + LoRA-adapted embeddings perform monument verification and confidence estimation. Successful verification unlocks historical context, travel metadata, and navigation support, transforming monument discovery into an interactive AI-assisted scavenger hunt experience.

## Classification Performance



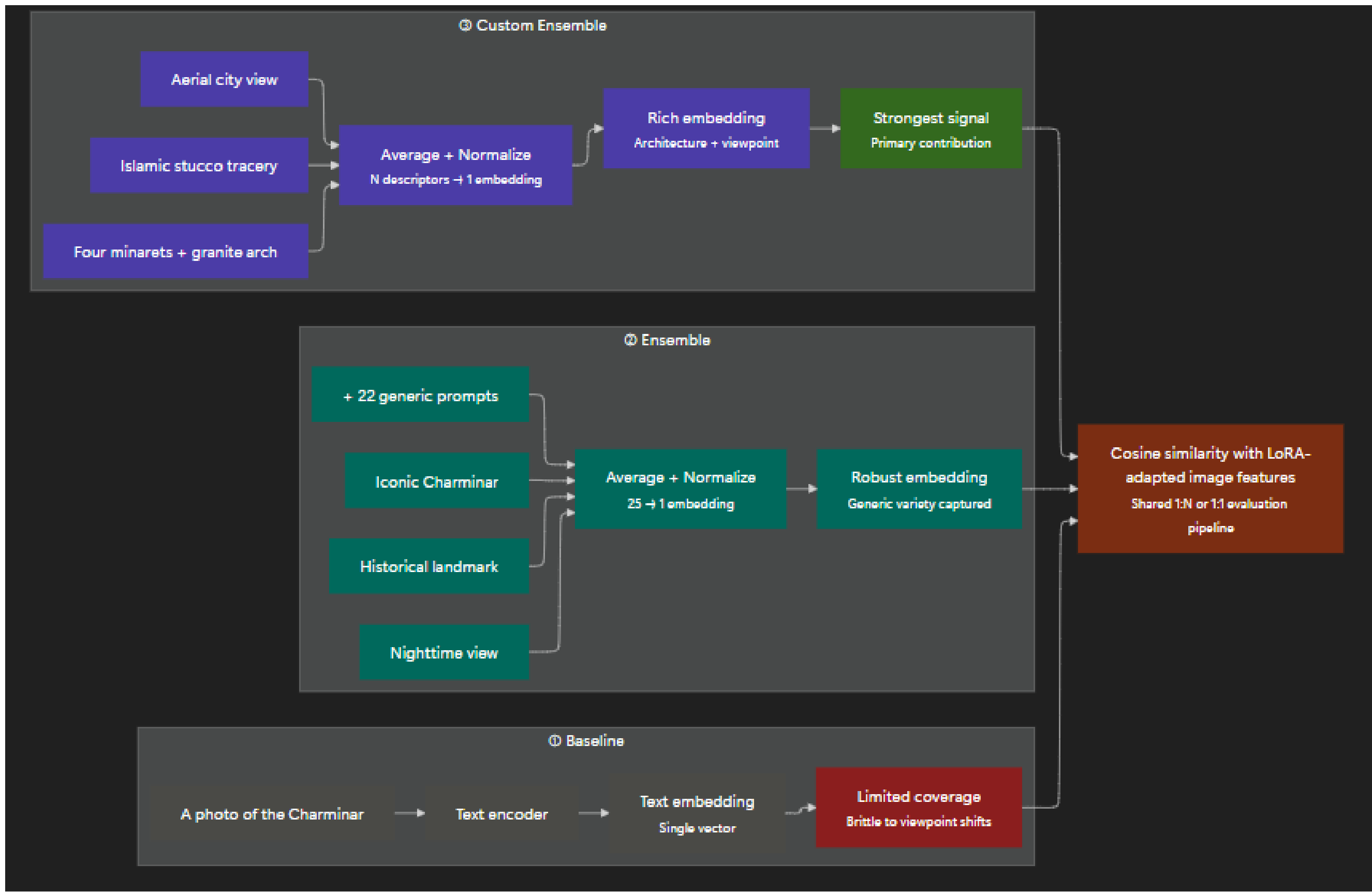
- Baseline CLIP achieved approximately 65–66% classification accuracy across prompt strategies.
- LoRA fine-tuning significantly improved monument discrimination performance, increasing accuracy to above 88%.
- The custom ensemble LoRA configuration achieved the highest accuracy of 92.34%.
- Results indicate that domain-specific representation adaptation and prompt engineering jointly improved robustness.

## Verification Metrics



- Equal Error Rate (EER) decreased substantially from approximately 7–8% in the baseline model to below 2% for LoRA-based approaches.
- ROC-AUC scores consistently exceeded 0.997 across fine-tuned configurations, indicating strong embedding separability.
- TAR @ 1% FAR improved from approximately 70% to over 97%, demonstrating strong verification reliability under strict false-acceptance constraints.
- Consistent improvements across EER, ROC-AUC, and TAR metrics validate the effectiveness of LoRA adaptation and ensemble prompting.

## Prompt Ensembling Strategy



- Ensemble prompting improves robustness by aggregating multiple textual viewpoints.
- Custom monument-specific descriptors produced the strongest verification performance.
- All prompt strategies share the same cosine-similarity evaluation pipeline using LoRA-adapted image embeddings.

## Failure Analysis



- Foreground-dominant selfies reduce monument visibility.
- Non-canonical architectural views lack discriminative features.
- Weather and lighting shifts introduce domain variance.

## Future Work

- Multi-language clue generation using LLM-based contextual prompting
- Knowledge distillation for efficient on-device monument verification
- Expanded monument coverage with larger geographically diverse datasets
- GPS-aware real-time scavenger hunt exploration workflows
- AR-assisted visualization for immersive heritage interaction